

EPISTEMIC WARRANT PROTOCOL

What an agent is justified in accepting — and why

EWP-0.2.0 · Policy reference-v2

MIT · Python 3.12+ · Canonical 14 · Pathological 12 · Hardening 26 · Invalid 35 · Goldens 26

EWP defines the deterministic boundary between what an AI agent's memory contains and what the agent is epistemically justified in accepting.

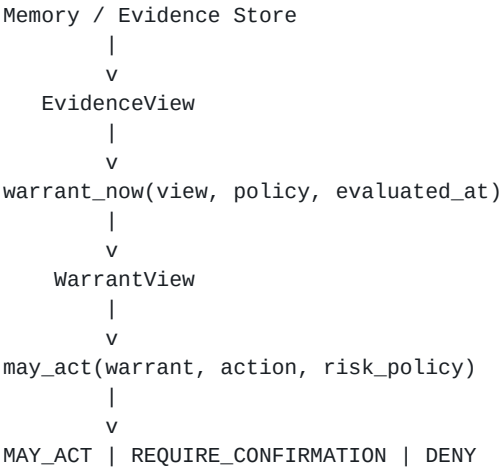
Memory is evidence, not truth.

Stores keep evidence. Warrant is computed. Action is a later gate.

Stores adapt to the protocol. The protocol does not inherit the store's epistemology.

1. What this is

EWP-0.2.0 is an epistemic protocol, not an action-authorization framework. Other projects use warrant to describe permission to act. EWP uses epistemic warrant to describe what an agent is justified in accepting. `may_act()` is deliberately a later gate.



Quick start

```
pip install .
ewp-ingest --db ./ewp.sqlite examples/quickstart.json --attest-trusted-origins
ewp-mcp --db ./ewp.sqlite

python3 tests/ci.py
python3 tests/report.py

from ewp.types import Policy
from ewp.warrant import warrant_now
from ewp.fixtures import fixture_verified_current, EVAL
print(warrant_now(fixture_verified_current(), Policy(), EVAL).warrant)
```

`pip install .` from a checkout installs the ewp package and the ewp-ingest and ewp-mcp commands; it is not on PyPI. `QUICKSTART.md` connects Claude to your own evidence. The repository is the reference implementation and the conformance suite.

2. Why this exists

Agents keep conversations, observations, documents, tool results, inferred facts, summaries, and prior decisions. Remembering something is not the same as knowing it is true. Most systems store and retrieve those records. EWP asks: given the evidence available at time T, under policy P, what may this agent accept — and why?

Common collapse	EWP split
stored = true	stored != believed
retrieved = important	retrieved != complete
repeated = corroborated	repeated != independently sourced
summarized = verified	summarized != externally checked
current = warranted	store-local current != warrant

believed = safe to act on	warranted != authorized to act
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The distinctions the protocol refuses to collapse.

3. The model

Evidence interchange

Stores expose assertions, evidence, provenance, lineage, conflicts, and verification records as an EvidenceView. Local current-fact, invalidation, ranking, and confidence machinery are inputs. They are not automatically agent beliefs.

Warrant evaluation

warrant_now(evidence_view, policy, evaluated_at) is deterministic. No I/O. No LLM. Same protocol version + same EvidenceView + same policy + same evaluated_at yields the same normative WarrantView. Invalid views are refused, not evaluated: values outside the closed enums, records about another proposition, duplicate or ambiguous ids, mistyped or missing fields. Warrant is a function, not a stored truth field.

Decision gating

An agent may be justified in accepting that a customer requested cancellation and still need confirmation before cancelling a \$2M contract. Risk, reversibility, authorization, privacy, money, and safety belong in action policy.

4. WarrantView axes

EWP does not collapse state into TRUE, VERIFIED, or DISPUTED. A proposition can be externally checked and still disputed; accepted and stale; well evidenced with an open conflict.

Axis	Values
acceptance	UNACCEPTED / TENTATIVE / ACCEPTED
conflict	NONE / OPEN / RESOLVED
verification	NONE / INDIRECT / EXTERNAL / HUMAN
currency	CURRENT / STALE / SUPERSEDED
sufficiency	SUFFICIENT / INSUFFICIENT / DEGRADED

Verification is a record (method, source, scope, time, result, freshness policy), not a badge. result=opposes opens conflict and blocks ACCEPTED. result=inconclusive cannot raise EXTERNAL or HUMAN. Endogenous origin caps human and external methods at INDIRECT. Identity is declared subjects[], compared as exact, case-insensitive ids: a check raises EXTERNAL or HUMAN only if neither side declares subjects or they share one. There is no family inference, and the scope string is never scraped. Currency is computed at evaluation time from checks that confer the chosen class. A later summary cannot refresh an old tool observation. History is not rewritten.

5. Lineage and the three confidences

Ten transformations of one source are not ten independent confirmations. The evaluator counts unique lineage_id values on assertions, evidence, and checks — not repetitions.

Kind	Meaning
Assertion confidence	How strongly the source or extractor stated the claim
Retrieval score	How relevant a record looks to this query
Warrant strength	How strongly policy permits acceptance

Implementations MUST NOT use retrieval relevance, repetition count, memory strength, or assertion confidence as substitutes for warrant.

6. Core invariants

- Assertions are not beliefs.
- Beliefs are not truth.
- Warrant is computed, not persisted as truth.
- Verification is evidence with method, scope, source, and time.
- Endogenous processing cannot manufacture external verification.
- Derivation cannot manufacture provenance.
- Contradiction must survive storage and retrieval.
- Incomplete retrieval must be visible as sufficiency=DEGRADED.
- Epistemic policy and action policy are separate.
- Semantically equivalent evidence must yield equivalent warrant independent of storage substrate.

Retrieval, summarization, dreaming, consolidation, reranking, repetition, graph propagation, LLM critique, and multi-agent agreement may reorganize. They may not turn verification=NONE into EXTERNAL without new external evidence. If search drops a contradictor, sufficiency is DEGRADED.

7. Platforms

OpenClaw

```
openclaw mcp add ewp --url http://127.0.0.1:8765/mcp
```

OpenClaw object	Role under EWP
Daily notes, transcripts	Raw lineage. Append. Do not rewrite.
USER.md	Working checkout of preferences. Regenerable.
MEMORY.md	Persona briefing from WarrantViews. Never authoritative.
Dreaming / promotion	Candidate generator. Must not raise verification class.
Local SQLite state	Cache. Not a second ledger.

Dreaming may rewrite MEMORY.md. EWP treats that rewrite as a new assertion, not as verification. Compression must not turn "X is disputed" into "X."

Claude, Codex, other MCP clients

Same contract. ewp-mcp (python3 -m ewp.mcp_server) speaks the MCP stdio transport (newline-delimited JSON-RPC). --http 127.0.0.1:8765 serves plain JSON-RPC POST /mcp for OpenClaw; it is not MCP Streamable HTTP. Every write requires the server-side ingest role (--allow-ingest on stdio, the token from EWP_INGEST_TOKEN on HTTP); without it the server is evaluate-only and opens an existing ledger read-only (create it with ewp-ingest). Trusted origins also need ingest_attestation=true. Inline views are hypothetical: ewp_warrant_now demotes their trusted origins and ewp_may_act refuses them. ewp_may_act evaluates the latest stored snapshot at server time. The pre-freeze claim/confidence sketch stays in historical/docs/MCP_CONTRACT.md and is not the shipped façade.

```
ewp-ingest --db ./ewp.sqlite evidence.json
ewp-mcp --db ./ewp.sqlite
EWP_INGEST_TOKEN=... ewp-mcp --http 127.0.0.1:8765 --db ./ewp.sqlite
openclaw mcp add ewp --url http://127.0.0.1:8765/mcp
```

Graphiti, Mem0, Particles, SQLite

Graphiti can be used as a temporal/entity evidence substrate or mirror. Inject lineage_id in episode metadata. invalid_at is store-local. valid_at is neither a verification check nor an observation time. Records belong to a proposition by the proposition_id their episodes declare, not by Graphiti group, and only the adapter's own group is read. Search collapse marks the view DEGRADED. Every fixture must round-trip through the fake Graphiti and Mem0 paths with identical axes. Live Graphiti and Mem0 mappings live in ewp/graphiti_client_adapter.py and ewp/mem0_adapter.py; notes in docs/implementer/LIVE_ADAPTERS.md. Live graphiti-core 0.30.2 is not validated. The Mem0 adapter targets the mem0ai 2.x clients and is tested against a real OSS mem0.Memory; every snapshot read is checked against the digest written at ingest. Mem0 default origin is extract; retrieval score is not warrant. Particles is a good immutable substrate; writes go only through EWP's ingest role. SQLite and JSON prove store neutrality.

Graphiti adapts to the protocol. The protocol does not adapt to Graphiti.

8. Why EWP sits above the store

System	Primary concern	EWP adds
Graphiti / Zep	Temporal knowledge graph and retrieval	Store-independent warrant evaluation
Mem0	Agent memory storage and retrieval	Evidence lineage and deterministic warrant policy
OpenClaw native memory	Inspectable agent context and memory	Separation of persona, evidence, and warrant
EWP	Epistemic evaluation	Not a general-purpose memory store

9. Conformance

```
python3 tests/ci.py
python3 tests/report.py
python3 tests/runner.py
python3 tests/runner_pathological.py
```

CI enforces the fixture, evaluator, policy, golden, and implementer-pack lock hashes; all 26 goldens; all 52 fixtures through the codec, SQLite, JSON, Mem0, and fake Graphiti with identical axes and fields (Graphiti's listed losses aside); 35 invalid views refused; the hardening pack; the Mem0 adapter against a real mem0ai 2.2.0 client; the MCP façade, including the official MCP SDK client; differential fuzz tests for evaluator parity, store neutrality, MCP arguments, and the JSON-RPC and HTTP transport; and an independent third evaluator. Changing a golden, the pack, the policy text, or the evaluator requires a protocol or policy version change, then `python3 tests/ci.py --write-lock`.

Class	Meaning
INGEST_LOSS	Store dropped assertions, sources, lineage, checks, or conflicts.
ADAPTER_MAP_LOSS	Store has the rows; EvidenceView is incomplete.
WARRANT_MISMATCH	Reconstructed view yields a WarrantView that is not the golden.
RETRIEVAL_LOSS	Search omitted evidence and did not mark DEGRADED.
EXPECTED_DIVERGENCE	Store current differs from warrant_now. Observation, not failure.

Pathological pack: false supersession, open contradiction, real temporal upgrade, same text / two lineages, three wordings / one lineage, store-current vs verified, weak invalidating strong, maintenance expiry that must not erase a check, retrieval-induced false consensus, extractor polarity flip on one lineage, canonical-without-verification, invalidation cycles.

10. Lock

```
Epistemic Warrant Protocol EWP-0.2.0
Policy: reference-v2
Canonical: 14/14   Pathological: 12/12   Hardening: 26/26   Invalid refused: 35/35
SQLite PASS   JSON PASS   Fake Graphiti PASS   Mem0 (fake client) PASS
graphiti-core 0.30.2 – NOT VALIDATED
fixture_set_sha256:
4358b59fdda94f522abbbb8fc810c1b344da7aebc1ba116bc3fa614e3bb6c419
evaluator_set_sha256:
fd446401b46ca19626cbdf287ecd6c67f7bf82c231e5135b92393774ae464e57
policy_set_sha256:
8a9f94fffc7fa95473232da5fb1de8120c26a70c7624eb126c67021412b6405b
golden_set_sha256:
```

```
1db7cab34639a87ce35c36635d4b6cffa46ba08855107fc9a04a4b89348a81c8
implementer_pack_sha256:
700153b13a020ce7f6abcb876a1b6fab86ec79b3b8931656fd1b051c0276fa0e
```

A store that produces a different answer has an adapter or conformance problem, not a license to move the goldens.

11. What EWP is not

Not a vector database, knowledge graph, memory engine, truth oracle, LLM fact-checker, or authorization framework. It does not ask what is ultimately true. It asks a narrower, computable question: given this bounded evidence view, at this time, under this versioned policy, what may the agent accept — and why?

EWP-0.2.0 binds verification to declared subjects[] as exact ids, scopes supersession to the proposition, treats future, missing, and unparsable times as unavailable at T, refuses invalid views, stores immutable snapshots, requires adapters to round-trip every field, and ships an MCP façade where every write needs the ingest role. The 26 golden axes are unchanged from 0.1.0. New stores may reveal adapter bugs; they do not redefine warrant.

Stores keep evidence. Warrant is computed. Action is a later gate.